```
import torch
import torch.nn as nn
import torch.optim as optim
import torchvision
import torchvision.transforms as transforms
from torch.utils.data import DataLoader
from torchvision import models, datasets
import matplotlib.pyplot as plt
import numpy as np
from sklearn.metrics import confusion_matrix, classification_report
import seaborn as sns
```

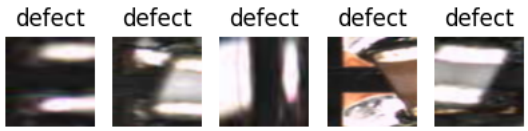
```
## Step 1: Load and Preprocess Data
# Define transformations for images
transform = transforms.Compose([
    transforms.Resize((224, 224)),  # Resize images for pre-trained model input
    transforms.ToTensor(),
    #transforms.Normalize([0.485, 0.456, 0.406], [0.229, 0.224, 0.225])  # Standard normalization for pre-trained models
])
```

```
!unzip -qq ./chip_data.zip -d data
```

```
dataset_path = "./data/dataset/"
train_dataset = datasets.ImageFolder(root=f"{dataset_path}/train", transform=transform)
test_dataset = datasets.ImageFolder(root=f"{dataset_path}/test", transform=transform)
```

```
def show_sample_images(dataset, num_images=5):
    fig, axes = plt.subplots(1, num_images, figsize=(5, 5))
    for i in range(num_images):
        image, label = dataset[i]
        image = image.permute(1, 2, 0)
        axes[i].imshow(image)
        axes[i].set_title(dataset.classes[label])
        axes[i].axis("off")
    plt.show()
```

```
show_sample_images(train_dataset)
```




```
print(f"Total number of training samples: {len(train_dataset)}")
first_image, label = train_dataset[0]
print(f"Shape of the first image: {first_image.shape}")
```

```
Total number of training samples: 172
Shape of the first image: torch.Size([3, 224, 224])
```

```
train_loader = DataLoader(train_dataset, batch_size=32, shuffle=True)
test_loader = DataLoader(test_dataset, batch_size=32, shuffle=False)
```

```
from torchvision.models import VGG19_Weights
model=models.vgg19(weights=VGG19_Weights.DEFAULT)
```

```
Downloading: "https://download.pytorch.org/models/vgg19-dcbb9e9d.pth" to /root/.cache/torch/hub/checkpoints/vgg19-dcbb9e9d.pth
100%|██████████| 548M/548M [00:07<00:00, 75.6MB/s]
```

```
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model = model.to(device)
```

```
from torchsummary import summary
summary(model, input_size=(3, 224, 224))
```

```
----------------------------------------------------------------
        Layer (type)               Output Shape         Param #
================================================================
            Conv2d-1         [-1, 64, 224, 224]           1,792
              ReLU-2         [-1, 64, 224, 224]               0
            Conv2d-3         [-1, 64, 224, 224]          36,928
              ReLU-4         [-1, 64, 224, 224]               0
         MaxPool2d-5         [-1, 64, 112, 112]               0
            Conv2d-6        [-1, 128, 112, 112]          73,856
              ReLU-7        [-1, 128, 112, 112]               0
            Conv2d-8        [-1, 128, 112, 112]         147,584
              ReLU-9        [-1, 128, 112, 112]               0
        MaxPool2d-10          [-1, 128, 56, 56]               0
           Conv2d-11          [-1, 256, 56, 56]         295,168
             ReLU-12          [-1, 256, 56, 56]               0
           Conv2d-13          [-1, 256, 56, 56]         590,080
             ReLU-14          [-1, 256, 56, 56]               0
           Conv2d-15          [-1, 256, 56, 56]         590,080
             ReLU-16          [-1, 256, 56, 56]               0
           Conv2d-17          [-1, 256, 56, 56]         590,080
             ReLU-18          [-1, 256, 56, 56]               0
        MaxPool2d-19          [-1, 256, 28, 28]               0
           Conv2d-20          [-1, 512, 28, 28]       1,180,160
             ReLU-21          [-1, 512, 28, 28]               0
           Conv2d-22          [-1, 512, 28, 28]       2,359,808
             ReLU-23          [-1, 512, 28, 28]               0
           Conv2d-24          [-1, 512, 28, 28]       2,359,808
             ReLU-25          [-1, 512, 28, 28]               0
           Conv2d-26          [-1, 512, 28, 28]       2,359,808
             ReLU-27          [-1, 512, 28, 28]               0
        MaxPool2d-28          [-1, 512, 14, 14]               0
           Conv2d-29          [-1, 512, 14, 14]       2,359,808
             ReLU-30          [-1, 512, 14, 14]               0
           Conv2d-31          [-1, 512, 14, 14]       2,359,808
             ReLU-32          [-1, 512, 14, 14]               0
           Conv2d-33          [-1, 512, 14, 14]       2,359,808
             ReLU-34          [-1, 512, 14, 14]               0
           Conv2d-35          [-1, 512, 14, 14]       2,359,808
             ReLU-36          [-1, 512, 14, 14]               0
        MaxPool2d-37            [-1, 512, 7, 7]               0
AdaptiveAvgPool2d-38            [-1, 512, 7, 7]               0
           Linear-39                 [-1, 4096]     102,764,544
             ReLU-40                 [-1, 4096]               0
          Dropout-41                 [-1, 4096]               0
           Linear-42                 [-1, 4096]      16,781,312
             ReLU-43                 [-1, 4096]               0
          Dropout-44                 [-1, 4096]               0
           Linear-45                 [-1, 1000]       4,097,000
================================================================
Total params: 143,667,240
Trainable params: 143,667,240
Non-trainable params: 0
----------------------------------------------------------------
Input size (MB): 0.57
Forward/backward pass size (MB): 238.69
Params size (MB): 548.05
Estimated Total Size (MB): 787.31
----------------------------------------------------------------
```

```
in_features = model.classifier[-1].in_features
model.classifier[-1] = nn.Linear(in_features, len(train_dataset.classes))
```

```
summary(model, input_size=(3, 224, 224))
```

```
----------------------------------------------------------------
        Layer (type)               Output Shape         Param #
================================================================
```

```
            Conv2d-1         [-1, 64, 224, 224]           1,792
              ReLU-2         [-1, 64, 224, 224]               0
            Conv2d-3         [-1, 64, 224, 224]          36,928
              ReLU-4         [-1, 64, 224, 224]               0
         MaxPool2d-5         [-1, 64, 112, 112]               0
            Conv2d-6        [-1, 128, 112, 112]          73,856
              ReLU-7        [-1, 128, 112, 112]               0
            Conv2d-8        [-1, 128, 112, 112]         147,584
              ReLU-9        [-1, 128, 112, 112]               0
        MaxPool2d-10          [-1, 128, 56, 56]               0
           Conv2d-11          [-1, 256, 56, 56]         295,168
             ReLU-12          [-1, 256, 56, 56]               0
           Conv2d-13          [-1, 256, 56, 56]         590,080
             ReLU-14          [-1, 256, 56, 56]               0
           Conv2d-15          [-1, 256, 56, 56]         590,080
             ReLU-16          [-1, 256, 56, 56]               0
           Conv2d-17          [-1, 256, 56, 56]         590,080
             ReLU-18          [-1, 256, 56, 56]               0
        MaxPool2d-19          [-1, 256, 28, 28]               0
           Conv2d-20          [-1, 512, 28, 28]       1,180,160
             ReLU-21          [-1, 512, 28, 28]               0
           Conv2d-22          [-1, 512, 28, 28]       2,359,808
             ReLU-23          [-1, 512, 28, 28]               0
           Conv2d-24          [-1, 512, 28, 28]       2,359,808
             ReLU-25          [-1, 512, 28, 28]               0
           Conv2d-26          [-1, 512, 28, 28]       2,359,808
             ReLU-27          [-1, 512, 28, 28]               0
        MaxPool2d-28          [-1, 512, 14, 14]               0
           Conv2d-29          [-1, 512, 14, 14]       2,359,808
             ReLU-30          [-1, 512, 14, 14]               0
           Conv2d-31          [-1, 512, 14, 14]       2,359,808
             ReLU-32          [-1, 512, 14, 14]               0
           Conv2d-33          [-1, 512, 14, 14]       2,359,808
             ReLU-34          [-1, 512, 14, 14]               0
           Conv2d-35          [-1, 512, 14, 14]       2,359,808
             ReLU-36          [-1, 512, 14, 14]               0
        MaxPool2d-37            [-1, 512, 7, 7]               0
AdaptiveAvgPool2d-38            [-1, 512, 7, 7]               0
           Linear-39                 [-1, 4096]     102,764,544
             ReLU-40                 [-1, 4096]               0
          Dropout-41                 [-1, 4096]               0
           Linear-42                 [-1, 4096]      16,781,312
             ReLU-43                 [-1, 4096]               0
          Dropout-44                 [-1, 4096]               0
           Linear-45                    [-1, 2]           8,194
================================================================
Total params: 139,578,434
Trainable params: 139,578,434
Non-trainable params: 0
----------------------------------------------------------------
Input size (MB): 0.57
Forward/backward pass size (MB): 238.68
Params size (MB): 532.45
Estimated Total Size (MB): 771.70
----------------------------------------------------------------
```

```
for param in model.features.parameters():
    param.requires_grad = False
```

```
# Include the Loss function and optimizer
# criterion =nn.BCEWithLogitsLoss()
# optimizer =torch.optim.Adam(model.parameters(), lr=0.001)
criterion = nn.CrossEntropyLoss()
optimizer = optim.Adam(model.parameters(), lr=0.001)
```

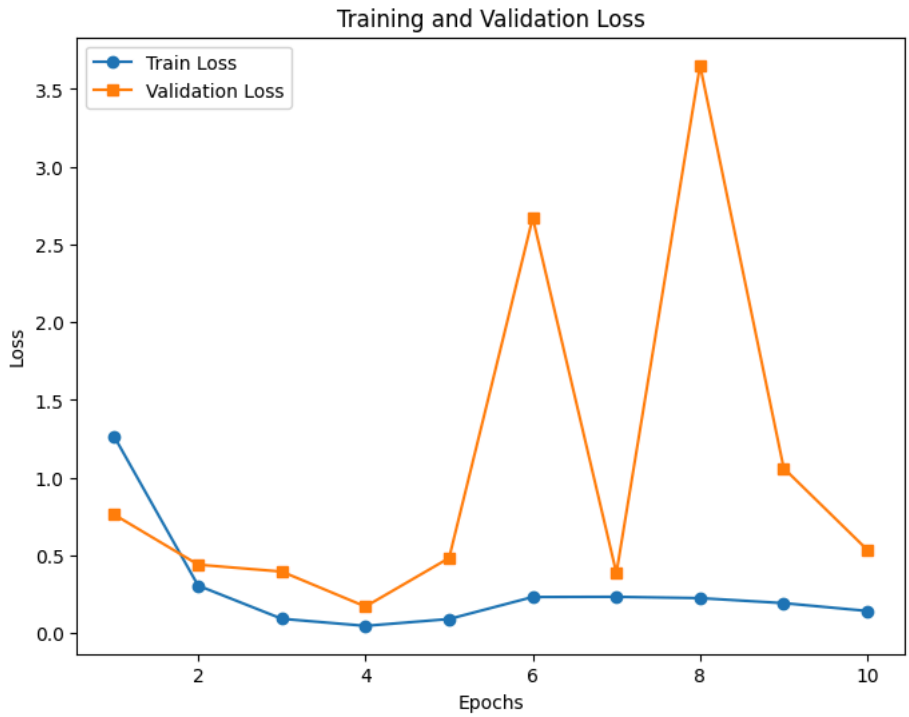
```
def train_model(model, train_loader,test_loader,num_epochs=10):
    train_losses = []
    val_losses = []
    model.train()
    for epoch in range(num_epochs):
        running_loss = 0.0
        for images, labels in train_loader:
            images, labels = images.to(device), labels.to(device)
            optimizer.zero_grad()
            outputs = model(images)
            loss = criterion(outputs, labels)
            loss.backward()
            optimizer.step()
```

```
            running_loss += loss.item()
        train_losses.append(running_loss / len(train_loader))

        # Compute validation loss
        model.eval()
        val_loss = 0.0
        with torch.no_grad():
            for images, labels in test_loader:
                images, labels = images.to(device), labels.to(device)
                outputs = model(images)
                loss = criterion(outputs, labels)
                val_loss += loss.item()

        val_losses.append(val_loss / len(test_loader))
        model.train()

        print(f'Epoch [{epoch+1}/{num_epochs}], Train Loss: {train_losses[-1]:.4f}, Validation Loss: {val_losses[-1]:.4f}')

    # Plot training and validation loss
    print("Name: KAAVIYAN K")
    print("Register Number:212224240066")
    plt.figure(figsize=(8, 6))
    plt.plot(range(1, num_epochs + 1), train_losses, label='Train Loss', marker='o')
    plt.plot(range(1, num_epochs + 1), val_losses, label='Validation Loss', marker='s')
    plt.xlabel('Epochs')
    plt.ylabel('Loss')
    plt.title('Training and Validation Loss')
    plt.legend()
    plt.show()
```

```
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model = model.to(device)
```

```
train_model(model,train_loader,test_loader,10)
```

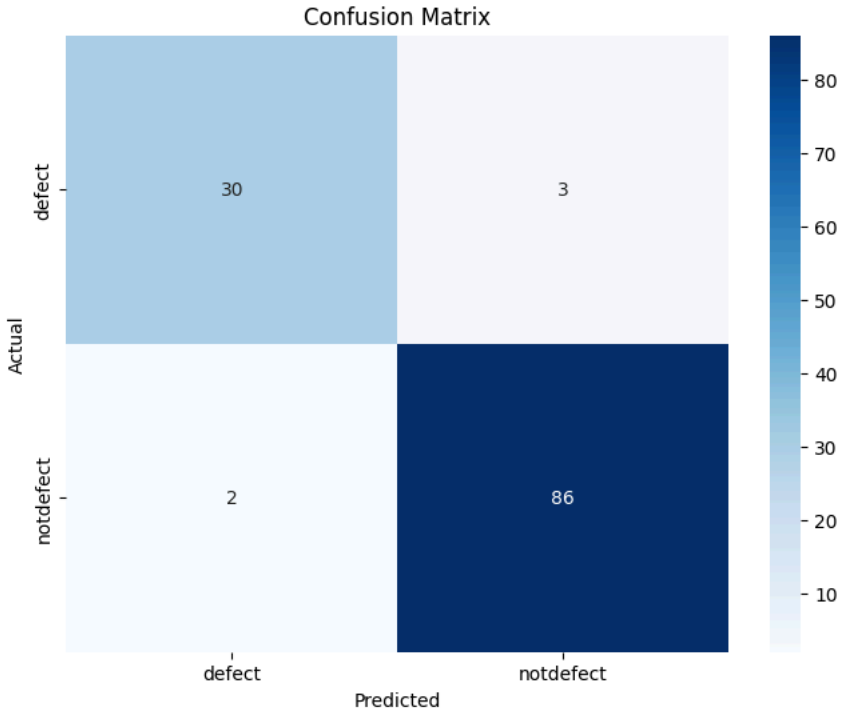
```
Epoch [1/10], Train Loss: 1.2613, Validation Loss: 0.7611
Epoch [2/10], Train Loss: 0.3056, Validation Loss: 0.4402
Epoch [3/10], Train Loss: 0.0919, Validation Loss: 0.3956
Epoch [4/10], Train Loss: 0.0464, Validation Loss: 0.1707
Epoch [5/10], Train Loss: 0.0905, Validation Loss: 0.4831
Epoch [6/10], Train Loss: 0.2322, Validation Loss: 2.6717
Epoch [7/10], Train Loss: 0.2334, Validation Loss: 0.3860
Epoch [8/10], Train Loss: 0.2250, Validation Loss: 3.6525
Epoch [9/10], Train Loss: 0.1927, Validation Loss: 1.0611
Epoch [10/10], Train Loss: 0.1423, Validation Loss: 0.5352
Name: KAAVIYAN K
Register Number:212224240066
```




```
def test_model(model, test_loader):
    model.eval()
    correct = 0
    total = 0
    all_preds = []
    all_labels = []

    with torch.no_grad():
        for images, labels in test_loader:
            images, labels = images.to(device), labels.to(device)
            outputs = model(images)
            _, predicted = torch.max(outputs, 1)
            total += labels.size(0)
            correct += (predicted == labels).sum().item()
            all_preds.extend(predicted.cpu().numpy())
            all_labels.extend(labels.cpu().numpy())

    accuracy = correct / total
    print(f'Test Accuracy: {accuracy:.4f}')

    # Compute confusion matrix
    cm = confusion_matrix(all_labels, all_preds)
    print("Name: KAAVIYAN K")
    print("Register Number:212224240066")
    plt.figure(figsize=(8, 6))
    sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=train_dataset.classes, yticklabels=train_dataset.classes)
    plt.xlabel('Predicted')
    plt.ylabel('Actual')
    plt.title('Confusion Matrix')
    plt.show()

    # Print classification report

    print("Classification Report:")
    print(classification_report(all_labels, all_preds, target_names=train_dataset.classes))
```

```
test_model(model,test_loader)
```

```
Test Accuracy: 0.9587
Name: KAAVIYAN K
Register Number:212224240066
```




```
Classification Report:
              precision    recall  f1-score   support

      defect       0.94      0.91      0.92        33
   notdefect       0.97      0.98      0.97        88

    accuracy                           0.96       121
   macro avg       0.95      0.94      0.95       121
weighted avg       0.96      0.96      0.96       121
```

```
def predict_image(model, image_index, dataset):
    model.eval()
    image, label = dataset[image_index]
    with torch.no_grad():
        image_tensor = image.unsqueeze(0).to(device)
        output = model(image_tensor)
        _, predicted = torch.max(output, 1)


    class_names = class_names = dataset.classes
    # Display the image
    image_to_display = transforms.ToPILImage()(image)
    plt.figure(figsize=(4, 4))
    plt.imshow(image_to_display)
    plt.title(f'Actual: {class_names[label]}\nPredicted: {class_names[predicted]}')
    plt.axis("off")
    plt.show()

    print(f'Actual: {class_names[label]}, Predicted: {class_names[predicted]}')
```

```
predict_image(model, image_index=25, dataset=test_dataset)
```

Actual: defect
Predicted: defect

```
Actual: defect, Predicted: defect
```

```
predict_image(model, image_index=55, dataset=test_dataset)
```

Could not connect to the reCAPTCHA service. Please check your internet connection and reload to get a reCAPTCHA challenge.